When case mix masquerades as batch drift: Prevalence-aware limits of unsupervised centering in breast cytology prediction

ChatGPT

Abstract

Background. Unlabeled feature-mean alignment can reverse synthetic measurement offsets, but a change in disease prevalence also changes the feature mean. These mechanisms may be confounded, potentially distorting both diagnostic predictions and their uncertainty sets. **Objective.** To quantify the harm caused by this confounding and evaluate a simple correction that preserves the class-separating direction. **Methods.** We reanalyzed 569 publicly available breast fine-needle aspiration cases with 30 image-derived nuclear features. In 60 reproducible, overlapping stratified data partitions, we fitted logistic regression, random forests, and histogram gradient boosting using separate calibration sets. We constructed unlabeled 60-case target batches with malignant prevalences of 20%, 40%, or 70% and applied five prespecified synthetic perturbation types. Three strategies were compared: no adaptation, full batch-mean centering, and projection of estimated drift away from the training-derived class-mean contrast. We evaluated balanced accuracy, AUROC, and 90% marginal or label-conditional conformal coverage. **Results.** In 70%-malignant batches without added measurement shift, ordinary centering reduced malignant label-conditional coverage to 62.3%, 65.1%, and 64.5% for logistic regression, random forest, and boosting. Axis-preserving alignment yielded 90.8%, 92.0%, and 91.7%, respectively. For logistic regression, the paired coverage difference between constrained and full centering was 28.5 percentage points (95% descriptive Monte Carlo interval, 26.8–30.2). Under nuisance-direction offsets, constrained alignment also improved coverage or balanced accuracy. Under shifts parallel to the class-separating direction, however, the correction could not restore coverage. **Conclusions.** Mean-based batch normalization can erase clinically relevant variation in class composition even when no assay change occurred. A label-free orthogonal correction avoids this specific mean-confounding mechanism, but its protection is structural and conditional, not a clinical validation or a guarantee under arbitrary dataset shift.

Keywords: breast cytology; batch effects; class prevalence; dataset shift; conformal prediction; calibration; transductive adaptation; computational pathology.

1 Introduction

Breast cytology prediction models often summarize a fine-needle aspirate through numerical properties of cell nuclei, such as size, contour, and texture. The Wisconsin Diagnostic Breast Cancer (WDBC) dataset has made this an accessible computational test bed, although it is a historical and single-institution cohort rather than a contemporary deployment dataset [Wolberg et al., 1993, 1994]. High discrimination on random partitions does not guarantee reliability if feature measurements, patient composition, or both change after deployment.

A familiar response to a technical batch effect is to shift target features so that the mean of an unlabeled target batch matches a reference mean. Under an exactly shared additive feature offset,

this is intuitively attractive. However, the mean feature vector also depends on the proportions of different biological phenotypes. If a clinic receives a larger fraction of malignant lesions, its mean cytology profile should change even with completely stable instrumentation. Matching target and reference means can then remove disease-associated information rather than nuisance variation.

This is connected to label shift, under which class proportions change while the within-class measurement distributions remain stable [Lipton et al., 2018]. It is also distinct from established batch adjustment for high-dimensional molecular assays, such as empirical-Bayes correction of microarray batches [Johnson et al., 2007]. We do not evaluate or claim to replace those methods. The practical question here is narrower: *when does unlabeled mean alignment in diagnostic cytology hide a genuine prevalence change, and can a geometric restriction protect the class signal?*

Uncertainty estimates make the problem more consequential. Conformal prediction produces finite-sample marginal coverage when calibration and future examples are exchangeable; a covariate or measurement shift can invalidate that premise [Angelopoulos and Bates, 2023, Tibshirani et al., 2019]. Even class-conditional conformal calibration, which can protect label-specific coverage under suitable within-class exchangeability, is not automatically robust to an adaptive preprocessing step that distorts those conditional distributions. Recent studies in medical image analysis emphasize the need to examine changes in annotation and population structure, not only image corruption [Boman et al., 2026].

We contribute three things: a transparent decomposition of observed mean drift into prevalence and measurement components; a deliberately simple *axis-preserving* negative-control correction, with an explicit identifiability limitation; and a reproducible, paired benchmark of three model classes, three prevalence settings, five perturbation mechanisms and two conformal variants. This is a method-focused stress test, not a new breast cancer biomarker or a claim that orthogonal centering is a universally novel algorithm.

2 Why unlabeled centering can fail

2.1 A population mean identity

Let $Z \in \mathbb{R}^p$ denote standardized image-derived features, $Y \in \{0, 1\}$ the benign/malignant label, $\mu_y = \mathbb{E}[Z \mid Y = y]$, and π_s and π_t the malignant proportions in the source reference and target populations. Suppose for the moment that the within-class feature distributions are unchanged except for a feature-wise additive offset b common to both target classes. Define the class contrast $d = \mu_1 - \mu_0$. Then

$$\underbrace{\mathbb{E}_t[Z] - \mathbb{E}_s[Z]}_{\Delta_{\text{mean}}} = \underbrace{b}_{\text{technical offset}} + \underbrace{(\pi_t - \pi_s)d}_{\text{case-mix component}} . \tag{1}$$

The identity follows directly by substituting $\mathbb{E}[Z] = (1 - \pi)\mu_0 + \pi\mu_1$. Crucially, no amount of target data can separate the two terms from *their means alone* if b is allowed to have an arbitrary component parallel to d. A decrease in π_t can be observationally indistinguishable, at the mean level, from an additive technical offset along d.

2.2 A constrained, label-free correction

Let $\hat{u} = \hat{d}/\|\hat{d}\|_2$ be the unit vector between the two source *training*-class feature means. For the calibration-set reference mean $\bar{z}_c$ and an unlabeled target-batch mean $\bar{z}_t$, compare three maps:

$$T_0(z) = z \qquad \text{(none)}, \tag{2}$$

$$T_{\text{full}}(z) = z - (\bar{z}_t - \bar{z}_c) \qquad \text{(full centering)}, \tag{3}$$

$$T_{\perp}(z) = z - (I - \hat{u}\hat{u}^{\top})(\bar{z}_t - \bar{z}_c) \qquad \text{(axis-preserving centering)}. \tag{4}$$

When $\hat{u}$ accurately estimates the population contrast direction, multiplication by $I - uu^{\top}$ eliminates the prevalence contribution in Equation 1. In the population limit,

$$(I - uu^{\top})\Delta_{\text{mean}} = (I - uu^{\top})b.$$

Thus the constrained adjustment can remove the component of an additive batch offset *orthogonal* to disease contrast without treating the corresponding case-mix change as nuisance. This is a geometric decomposition, not a new general conformal coverage theorem.

Its limitation is equally important. If b is parallel to d, the constrained map cannot correct that technical shift. Fully correcting b and completely protecting an unknown change in π are generally incompatible when only the unlabeled means and unrestricted additive b are observed. Finite-sample errors in $\hat{d}$ and batch means, class-specific measurement artifacts, nonlinear shifts, and changing $P(Y \mid Z)$ create additional failure modes.

3 Materials and methods

3.1 Data, models and separation of roles

WDBC provides 569 labeled cases (212 malignant, 357 benign) and 30 continuous nuclear-morphology measurements. We used the frozen data distributed with scikit-learn, not downloaded secondary reconstructions [Wolberg et al., 1993, Pedregosa et al., 2011]. For each of 60 predetermined integer seeds (0–59), we randomly stratified cases into approximately 60% model training (341), 20% labeled conformal calibration (114) and 20% test pool (114). The same people necessarily recur across seeds; the 60 repeats are correlated computational resamples, *not* 60 independent hospitals or populations [Varoquaux, 2018].

Training-only standardization preceded three fixed models: ℓ_2-penalized logistic regression ($C = 1$), a 120-tree random forest (minimum leaf size two) and histogram gradient boosting (120 iterations, maximum nine leaves, learning rate 0.05, ℓ_2 regularization 1). No model or correction was selected by held-out diagnostic results. The reference centroid used only calibration *features*, while the class axis used only training *labels and features*. The calibration labels set conformal score thresholds. Outcomes in the test pool were used solely to construct the simulated class-mixture scenarios and to evaluate results; such prevalence construction is an experimental intervention, not information available to the adaptation algorithm.

3.2 Target batches and shift mechanisms

For each seed we drew separate, non-replacement samples of 60 test cases with malignant fractions $\pi_t = 0.20$, 0.40, or 0.70 (12, 24 or 42 malignant cases). These fractions represent controlled

case-mix sensitivity analyses, not empirical prevalence at any clinic. Each batch was exposed to five conditions in standardized training-feature units:

1. **No technical shift:** $Z' = Z$, a negative control for prevalence confounding.
2. **Orthogonal nuisance offset:** $Z' = Z + b_\perp$ with $b_\perp \perp \hat{u}$ and $\|b_\perp\|_2 = 0.75\sqrt{p}$.
3. **Mixed shared offset:** $Z' = Z + b$ for isotropic Gaussian direction b, rescaled to norm $0.75\sqrt{p}$.
4. **Disease-axis offset:** $Z' = Z + b_\parallel$ with $b_\parallel \parallel \hat{u}$, norm $0.75\sqrt{p}$ and randomly selected sign; a deliberate negative control for the constrained method.
5. **Case-specific noise:** $Z'_{ij} = Z_{ij} + 0.50\epsilon_{ij}$, $\epsilon_{ij} \overset{\text{iid}}{\sim} N(0,1)$; centering cannot remove individual-case corruption.

Shared offsets were fixed across all 60 cases of a batch, and new random realizations were drawn across seeds and scenarios. Each of the three frozen models received the same shifted batches and adaptation methods within the same seed, permitting paired comparisons. There were $60 \times 3 \times 3 \times 5 \times 3 \times 2 = 16{,}200$ model–batch–method–conformal evaluations. The perturbations are synthetic; their magnitudes were *not* derived from observed microscope, staining, or site-to-site error distributions.

3.3 Conformal sets and prespecified outcomes

For a frozen classifier producing $\hat{p}(y \mid z)$, each calibration example receives score $S_i = 1 - \hat{p}(Y_i \mid Z_i)$. With m calibration scores, the marginal threshold is their $\lceil (m+1)(1-\alpha) \rceil$-th smallest value (or $+\infty$ if this rank exceeds m). At $\alpha = 0.1$, the prediction set is

$$\widehat{C}(z) = \{y \in \{0,1\} : 1 - \hat{p}(y \mid z) \leq q\}.$$

Our *label-conditional* variant computes separate thresholds q_y among calibration cases with label y; this is a Mondrian-style label partition, not the adaptive score construction of Romano et al. [2020]. Both calibration schemes used unchanged source scores. Exchangeability supports their corresponding marginal or within-label coverage statements *before* unmodeled distribution shifts; no postadaptation coverage guarantee is claimed.

The **primary descriptive endpoint** was observed coverage for the malignant label under label-conditional conformal calibration, reported with the benign-label coverage as a companion endpoint. We also report empirical overall coverage, mean prediction-set size, balanced accuracy, AUROC and Brier score. Nominal 90% is a reference line, not a clinical certification threshold. Pairwise differences were computed within identical seed and target batch. Percentile intervals are bootstrap intervals over the 60 repeated seed-level evaluations; because individuals recur, these describe *Monte Carlo variability* of this resampling design and are not independent-patient or external-validation confidence intervals. We avoid inferential p-values for clinical generalization.

4 Results

4.1 Prevalence change alone can make centering harmful

Figure 1 isolates the central contrast: the target class mix is changed by resampling, while the feature measurements are left untouched. At 70% malignant cases, ordinary centering lowered

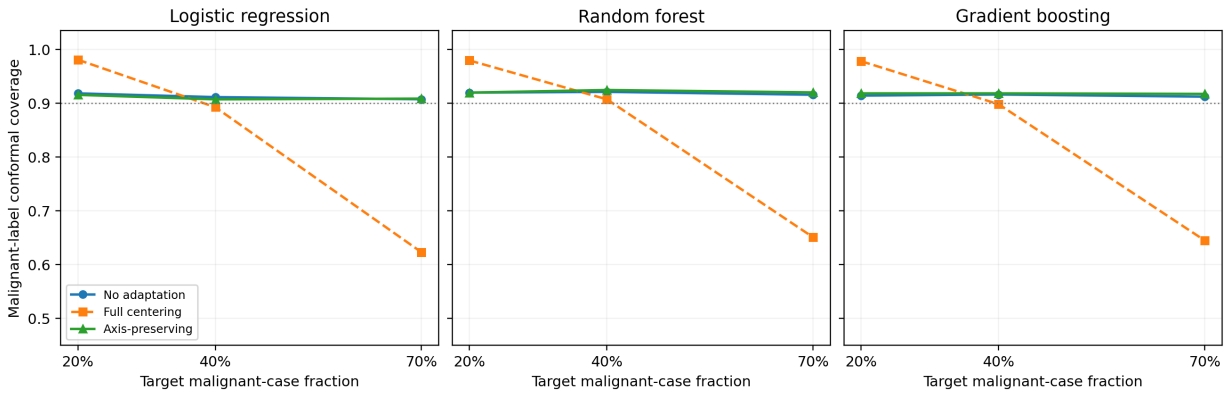


Figure 1: Changing class mixture without altering measurements. Observed malignant-label coverage at 90% class-conditional nominal coverage, across 60 overlapping resamples. The gray dotted line marks 90%. Case mix alone can cause full mean centering to change classification and conformal reliability.

Table 1: Malignant-label 90% conformal coverage in constructed 70%-malignant target batches. All values are means over 60 repeated WDBC splits, not external-cohort estimates. Higher coverage alone is not sufficient to establish more informative sets.

Model	Shift	No adaptation	Full centering	Axis-preserving
Logistic	None	.907	.623	.908
	Orthogonal	.903	.623	.908
	Mixed	.890	.623	.899
	Disease axis	.715	.623	.715
Forest	None	.915	.651	.920
	Orthogonal	.884	.651	.920
	Mixed	.879	.651	.908
	Disease axis	.727	.651	.726
Boosting	None	.912	.645	.917
	Orthogonal	.864	.645	.917
	Mixed	.860	.645	.899
	Disease axis	.722	.645	.723

malignant class-conditional coverage to 0.623 in logistic regression, 0.651 in random forest and 0.645 in boosting (Table 1). Without adaptation, the respective coverages were 0.907, 0.915, and 0.912. Axis-preserving centering retained 0.908, 0.920, and 0.917.

For logistic regression, the paired axis-preserving-minus-full-centering difference was +0.285 (95% descriptive Monte Carlo interval +0.268 to +0.302). The analogous differences were +0.269 (0.252–0.285) for forest and +0.272 (0.256–0.287) for boosting. These differences reflect the engineered resampling interventions on the WDBC cohort, not transportability to another clinic.

The apparent coverage failure is class-asymmetric. With 70% malignancy and no measurement shift, full centering produced benign coverage 1.000 but malignant coverage 0.623 for logistic regression. Its balanced accuracy was 0.844, versus 0.969 with no correction. Conversely, in 20%-malignant batches, full centering attained malignant coverage 0.981 while benign coverage fell to 0.654. A single overall coverage statistic would obscure this trade-off.

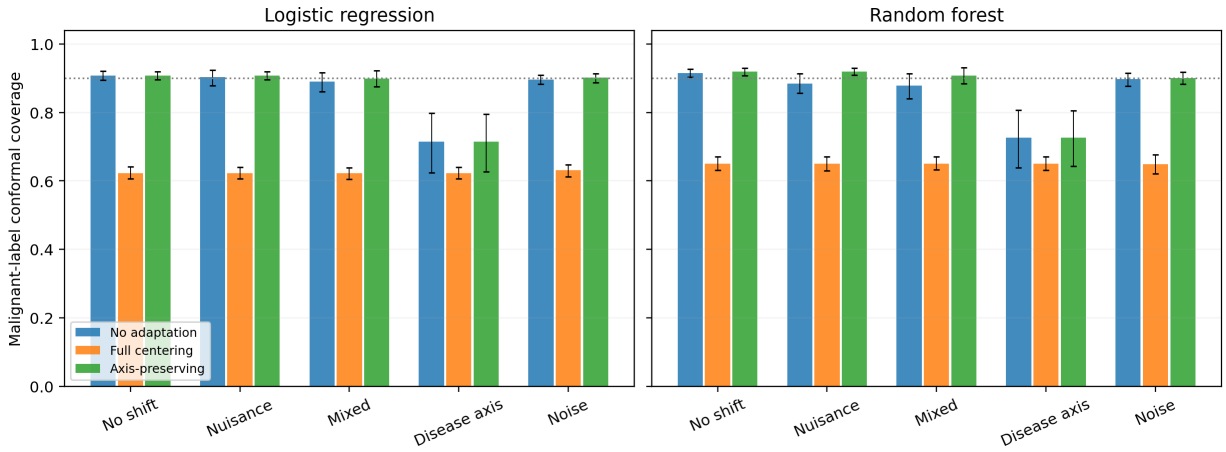


Figure 2: Failure analysis in high-malignancy test batches. Malignant-label conformal coverage at target prevalence 70% under five engineered mechanisms. Bars show split means; whiskers are descriptive 95% Monte Carlo resampling intervals, not clinical confidence limits. The axis-preserving method does not correct disease-axis offsets.

4.2 Recovery is mechanism-specific, not universal

For 70%-malignant batches with a shared orthogonal nuisance offset, the constrained method recovered malignant coverage of 0.908, 0.920 and 0.917 across the three models, compared with 0.903, 0.884 and 0.864 for no correction. Its balanced accuracy for these models was 0.968, 0.948 and 0.954; corresponding no-correction balanced accuracies were 0.948, 0.927 and 0.894. The contrast supports its intended *nuisance-axis* role (Figure 2).

The negative controls are at least as important. Under large offsets *parallel* to the disease axis, axis-preserving centering gave malignant coverage only 0.715, 0.726 and 0.723. This matches the predicted inability to distinguish a disease-aligned technical offset from a genuine prevalence contribution using the batch mean. Case-specific Gaussian noise also could not be exactly canceled by any mean adjustment. It would therefore be incorrect to infer a blanket 90% coverage guarantee from the favorable nuisance-axis scenarios.

Full centering returns the same transformed values under all *shared* additive offsets applied to a fixed batch (the offset cancels exactly). Accordingly its values are identical across the "none," orthogonal, mixed and disease-axis rows of Table 1. That consistency is a consequence of construction, not evidence of biological robustness.

4.3 Smaller-batch sensitivity analysis

As a separate, lower-precision sensitivity check, we repeated the same factorial study on 30-case target batches for the first 20 seeds, and compared these with the matched first 20 seeds of the 60-case analysis. Under orthogonal shifts and 70%-malignant case mix, the axis-preserving malignant coverages at target batch sizes 30 versus 60 were 0.914 versus 0.914 (logistic), 0.926 versus 0.924 (forest), and 0.919 versus 0.915 (boosting). However, under 20% malignancy, the boosting coverage changed from 0.850 (30 cases) to 0.913 (60 cases), highlighting instability when the target batch is small. These are descriptive checks on overlapping case pools, not evidence of a universal minimum batch size. Detailed benign and malignant coverage values are in `supplement/batch_size_sensitivity.csv`.

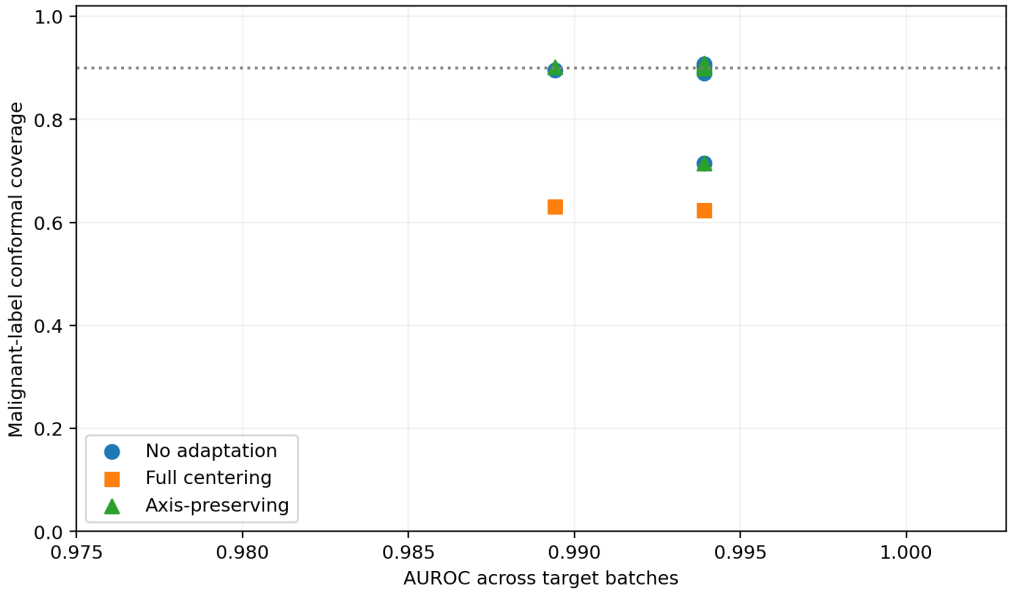


Figure 3: Discrimination and malignant coverage answer different questions. Logistic regression across the five technical-shift conditions at malignant prevalence 70%. Points correspond to summary values across 60 resamples, with three adaptation strategies. AUROC remains high even when malignant-label coverage is low.

4.4 AUROC does not resolve uncertainty failure

On prevalence-altered but unperturbed test batches, logistic regression maintained an AUROC near 0.99 despite coverage varying widely across normalization strategies. Adding a constant standardized feature offset to every logistic-regression input adds a constant to all model logits. Consequently pairwise score order and AUROC remain invariant under such an intervention, while class probabilities, set membership and fixed-threshold accuracy can change. Figure 3 illustrates this divergence. The stronger claim is mechanistic rather than clinical: high ranking performance does not by itself establish a valid or useful prediction set.

5 Discussion

The study isolates a concrete failure mechanism for unlabeled batch normalization: a feature-centroid shift cannot generally distinguish changes in measurement from changes in phenotype frequency. In our controlled reanalysis, full centering moved prediction scores in a class-dependent way, sharply reducing malignant-label coverage when malignant prevalence increased. The proposed axis-preserving map is a narrow, interpretable guardrail: it explicitly refuses to correct target shifts along the source class-contrast direction. This is why it can protect signal under a pure prevalence intervention, and why it necessarily fails under disease-axis technical distortion.

The result does not establish that a particular laboratory should use the constrained correction. Cytology measurements can experience complex staining, microscope, slide-preparation, delineation, and label-adjudication variation. Those mechanisms need not be additive, linear, or orthogonal to diagnostic signals. The chosen class direction may also be unstable when features are correlated, as they are in WDBC, or if the future phenotype spectrum differs materially from source cases. Clinical usefulness requires outcomes, annotation protocols, and equipment histories from genuinely new acquisition sites.

This study should also not be conflated with methods designed to estimate label priors from unlabeled classifier outputs [Lipton et al., 2018] or weighted conformal prediction under identified covariate shift [Tibshirani et al., 2019]. Those methods answer related but distinct questions and have their own assumptions. Our restricted centering strategy neither estimates π_t nor calibrates a new target probability model. Compared with the extensive literature on conformal prediction and medical dataset shift, its contribution is chiefly a *transparent demonstration of a specific identifiability trap and its geometric trade-off*, rather than a new general-purpose domain-adaptation theory.

5.1 Strengths, limitations and publication scope

Strengths include a deliberately paired, reproducible design; multiple prediction architectures; separate class-specific coverage analyses; five failure-oriented perturbations; an explicit mean-shift identity; and independently runnable source, outputs and unit tests. The benchmark includes both a favorable mechanistic control (nuisance-orthogonal shift) and an unfavorable one (disease-aligned shift), reducing the risk of presenting only successful adaptation settings.

The most consequential limitation is **lack of independent external validation**. The WDBC cases are historical and from one institution. All repeated splits reuse the same people and cannot justify independent clinical confidence intervals. The shifts are fully synthetic and use user-chosen standardized magnitudes, not empirically estimated technical noise. No prospective workflow, age/subtype stratification, outcome costs, reader variability, or heterogeneous-site adaptation was measured. The target batches contain 60 cases, so transductive adjustment assumes simultaneous access to a substantial unlabeled cohort; single-patient deployment was not studied. With $p = 30$ and 341 training cases per split, the class-mean axis can be sensitive to sampling noise. We did not optimize hyperparameters, perform a complete comparative domain-adaptation review, or benchmark on an unrelated external hospital population.

These limitations matter for journal selection. Collins et al. [2024] emphasizes transparent description of evaluation populations, missing information and model validation. A methodological simulation with one public historical cytology cohort can be submitted as an exploratory methods or reproducibility article if it fits a journal's aims, but it should *not* be represented as a clinically validated diagnostic intervention. Q2–Q3 ranking and editorial suitability are journal- and year-specific and are not independently established by this manuscript.

6 Conclusion

Our experiment demonstrates that unlabeled centroid matching may erase disease-composition information and compromise malignant-case uncertainty coverage, even in the absence of technical corruption. Projection away from a training-derived class-contrast axis avoids this particular prevalence-confounding problem and corrects only its compatible nuisance directions. The informative negative result is the nonidentifiability of a technical offset aligned with the disease axis from batch means alone. The next defensible step before clinical or stronger methodological claims is evaluation on an independent, annotated acquisition cohort with measured assay variation and heterogeneity.

Reproducibility and declarations

Data and code. The study uses the public UCI WDBC dataset via `load_breast_cancer()`. A dataset SHA-256 fingerprint, complete script, scenario-level output table, figures, exact random seeds, pinned package versions and test suite are included in the accompanying archive. The scripts require no remote data download and reproduce both the successful and unsuccessful scenarios.

Ethical scope. This is a secondary computational analysis of a public de-identified benchmark. No new patient contact, new clinical intervention, or prospective evaluation occurred. Institutional ethics determinations and clinical deployment permissions are not asserted.

Funding, conflicts and author contributions. To be completed by the actual submitting authors. An automated analysis/drafting system assisted manuscript and code preparation; human authors must review, validate, and take responsibility for the final manuscript and make disclosures required by the target journal.

Availability statement. All evaluation outputs are provided locally in the supplemental archive. A persistent public repository and DOI should be established before journal submission.

Supplementary methodological notes

Exact quantities. For each seed, models were fit once, conformal cutoffs determined once on the unaffected calibration subset, and each target transformation applied without using target labels. A classwise threshold for each of the two labels was calculated independently. The same target selected cases, corruption vector and random seed were used across the models and adaptation arms. The complete factorial comparison includes 16,200 scenario-level model evaluations.

Interpretation of Monte Carlo bands. Percentile bands are from resampling the 60 deterministic-seed outputs; they measure precision of the scenario averages under the chosen randomized split design and must not be read as population-level intervals. Patient-level dependence across repeated splits precludes that interpretation. Exact conditional 90% guarantees do not automatically extend to adaptively transformed target features.

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